

the ARMA volume forecasting model improvement over the naïve thirty day average daily volume measure (e.g., ADV30). The error between actual volume and forecasting volume decreases -19.5% with the ARMA model over the naïve ADV30 measure.

Author's Note:

- In theory, the beta term in an ARMA model is often shown to be forecasted without the constant term alpha, but since we are using a moving median it is not guaranteed that the mean error will be zero and thus a constant term is needed.
- The ARMA forecast with the “beta” autoregressive terms can be computed both with and without special event days. Since it is important that this technique be continuous, unlike the day of week adjustment, we need to include all days. As an adjustment, we can: (1) Treat the special event day and day after the special event as any other day and include an adjustment for the previous day's forecasted error, (2) define the forecast error to be zero on a special event day (this way it will not be included in the next day's forecast), or (3) use a dummy variable for special event days.
- Our analysis calculated an autoregressive term across all stocks in each market cap category. Users may also prefer to use a stock specific autoregressive term instead. We did not find statistical evidence that a stock specific beta is more accurate for large cap stocks but there was some evidence supporting the need for a stock specific beta for the small cap stocks. Readers are encouraged to experiment with stock specific forecasts to determine what works best for their specific needs.

FORECASTING MONTHLY VOLUMES

In this section we describe a process to forecast average monthly volume levels. This process could also be extended to estimate annual volume levels. Having a forward looking ADV estimate can be very helpful for the portfolio manager who is looking to rebalance his/her portfolio at some future point in time when volumes may look much different than they do now.

Methodology:

Period: Ten years of data: Jan. 2002 through Dec. 2011.

Universe: Average daily volumes for large cap (SP500) and small cap (R2000) stocks by month.